

Coffee husk waste as a carrier for bioagents: development of *Trichoderma-Bacillus* consortia and their antimicrobials against soil borne *Phytophthora cinnamomi* Rands. in avocado

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Highlights

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[Trichoderma harzianum](#) T(MP)-7 and [Bacillus amyloliquefaciens](#) B(TI)-3 demonstrated strong antagonistic activity against *Phytophthora cinnamomi* PC(TKI)-10.

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Coffee husk-based bioformulation maintained high microbial viability and significantly enhanced avocado [seedling growth](#).

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GC-MS analysis identified multiple bioactive antimicrobial metabolites in the coffee husk-based bioformulation.

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Antifungal gene presence was confirmed in *T. harzianum* T(MP)-7 and *B. amyloliquefaciens* B(TI)-3.

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Coffee husk-based bioformulation effectively reduced *P. cinnamomi* disease severity in pot experiments.

Abstract

In vitro studies identified [Trichoderma harzianum](#) T(MP)-7 and [Bacillus amyloliquefaciens](#) B(TI)-3 as effective biocontrol agents against [Phytophthora cinnamomi](#) PC(TKI)-10. The isolates were found compatible under *in vitro* conditions, and a bioformulation combining them in a 1:1 ratio was tested in different substrates. Among tested, coffee husk-based bioformulations supported the highest microbial populations, with initial counts of *T. harzianum* T(MP)-7 at 7.5×10^{12} CFU/g and *B. amyloliquefaciens* B(TI)-3 at 6.8×10^{11} CFU/g, which declined to 3.1×10^7 and 3.8×10^8 CFU/g, respectively, by day 120. In soil amended with coffee husk-based bioformulations, initial counts of *T. harzianum* T(MP)-7 (5.8×10^7 CFU/g) and *B. amyloliquefaciens* B(TI)-3 (4.3×10^7 CFU/g) declined to 2.4×10^3 and 1.5×10^3 CFU/g, respectively, over the same period. Coffee husk-based bioformulation significantly enhanced avocado [seedling growth](#), achieving a 92 % germination rate, 14.00 cm root length, 52.43 cm shoot length, and an average of 16.33 leaves. GC-MS analysis identified multiple [antifungal](#) compounds, including [azelaic acid](#), [palmitic acid](#), [stearic acid](#), [squalene](#), [caffeic acid](#), [gallic acid](#), [chlorogenic acid](#), and 3-hydroxybenzoic acid. Additionally, *B. amyloliquefaciens* B(TI)-3 harbored [antifungal](#) antibiotic-producing genes, successfully amplifying [iturin](#) D (800 bp), bacillomycin D (482 bp), [iturin](#) C (506 bp), bacillomycin A (344 bp), fengycin D (220 bp), mycosubtilin (482 bp), [surfactin](#) (675 bp), and β -glucanase (400 bp). Similarly, *T. harzianum* T(MP)-7 tested positive for [chitinase](#) (1700 bp) and β -1,3-glucanase (850 bp). Soil application of coffee husk-based bioformulation at 10 g/kg, alone or combined with [arbuscular mycorrhizal fungi](#) (AMF) and [micronutrients](#) significantly reduced *P. cinnamomi* disease severity. The combined treatment resulted in a mean disease severity index (DSI) of 16.00 %, representing a 64.71 % reduction over control, while coffee husk-based bioformulation alone reduced DSI to 21.31 % (53.87 % reduction). These findings highlight the potential of coffee husk-based bioformulations as a sustainable biocontrol strategy against *P. cinnamomi*, improving plant health and growth in avocado cultivation.

Introduction

Avocado (*Persea americana* Mill.) is a globally significant fruit crop valued for its nutritional and commercial importance [1]. However, its production is severely threatened by *Phytophthora cinnamomi*, a highly destructive soilborne oomycete that causes root rot [2,3]. This pathogen thrives in high-humidity and poorly drained soils, leading to substantial yield losses and highlighting the need for effective and sustainable disease management strategies [4]. Conventional control methods, such as chemical fungicides, have limited long-term efficacy due to environmental concerns and the emergence of fungicide-resistant strains [5]. Consequently, there is an urgent need to explore eco-friendly

alternatives, particularly biocontrol-based strategies, for managing *P. cinnamomi* infections in avocado.

Among the most promising biocontrol agents, *Trichoderma* spp. and *Bacillus* spp. exhibit strong antagonistic properties against plant pathogens through mechanisms such as mycoparasitism, competition for nutrients, and the production of antimicrobial metabolites [6]. Additionally, these microorganisms promote plant growth by enhancing nutrient uptake and stimulating plant defense responses [7]. The co-cultivation of *Bacillus* and *Trichoderma* has been shown to enhance secondary metabolite production and pathogen suppression, yielding more bioactive compounds than when cultured separately, making them highly effective for large-scale agriculture [8]. Pot experiments have demonstrated that their combined application improves root and shoot biomass, nutrient absorption, and resistance to *Phytophthora* and *Fusarium* [9]. Similarly, *T. viride* and *B. subtilis* have been reported to reduce disease incidence in rice while enhancing grain yield [10,11]. Furthermore, *T. harzianum* application in carrot crop was found to boost beneficial microbial populations, such as *Pseudomonas* and *Bacillus* [11]. These findings underscore the potential of microbial consortia in improving crop health and disease resistance [9,12].

The efficacy of biocontrol agents can be significantly enhanced through carrier-based formulations that support microbial viability and sustained metabolite production [7,13]. Microbial bio-inoculant formulations include liquid (cell suspensions), solid (peat, charcoal, vermiculite), polymeric (agar, alginate, chitosan), and metabolite-based preparations [14]. Among potential carriers, coffee husk, a major agro-industrial byproduct, has emerged as a promising substrate for bioformulations due to its rich organic composition, biodegradability, and ability to enhance microbial activity [15]. Recent studies have shown that coffee husk derived metabolites exhibit potent antifungal properties, effectively inhibiting microbes [16]. The mode of action involves the disruption of fungal cell walls and membranes, mediated by bioactive compounds such as chlorogenic acid, caffeic acid, catechins, and quercetin [17].

Several secondary metabolites produced by *T. harzianum* and *B. amyloliquefaciens*, including lipopeptides, polyketides, and volatile organic compounds, have demonstrated strong antifungal and oomycetocidal activity [8]. However, research on the production of these metabolites in coffee husk-based bioformulations and their role in suppressing *P. cinnamomi* in avocado remains largely unexplored. Understanding the bioactive compounds involved could facilitate the development of novel biocontrol formulations with improved efficacy, stability, and field applicability. This study focuses on the evaluation and standardization of substrates, identification of antimicrobial metabolites

from coffee husk-based bioformulations, and assessment of their effectiveness against *P. cinnamomi*.

Source of materials used in the study

Ten endophytic bacterial isolates from avocado, designated as B(PI)-1 to B(TD)-10, and ten *Trichoderma* spp., labeled T(PI)-1 to T(TD)-10, were isolated. Based on *in vitro* studies, *Trichoderma harzianum* T(MP)-7 (NCBI Accession No: OR991815) and *Bacillus amyloliquefaciens* B(TI)-3 (NCBI Accession No: OR462348) were identified as effective against *P. cinnamomi* PC(TKI)-10 (NCBI Accession No: OR991996.1) using a dual-plate assay. These bioagents exhibited significant inhibitory effects, with *T.*

Testing compatibility between biocontrol agent

The compatibility of *T. harzianum* T(MP)-7 and *B. amyloliquefaciens* B(TI)-3 was tested using three different methods. In the paired plate assay, both microorganisms were cultured at equal distances from each other. There was no visible inhibition and eventually, *Trichoderma* grew over the *Bacillus* colony. In the PDA well diffusion assay, mycelial bit of *Trichoderma* was placed at the center of the plate while *Bacillus* was added to the surrounding wells. *Bacillus* did not inhibit the growth of *T.*

Discussion

The compatibility assays between *T. harzianum* T(MP)-7 and *B. amyloliquefaciens* B(TI)-3 showed no growth inhibition, indicating strong compatibility. *Trichoderma* grew alongside or over *Bacillus*, supporting their co-application in disease management. Similar findings by Izquierdo-García et al., [8] demonstrated that *T. virens* Gl006 and *B. velezensis* Bs006 effectively controlled *Fusarium* wilt without interfering with each other's antagonistic activity. Sarma et al., [24] reported that

Conclusion

This study demonstrates the potential of a coffee husk-based bioformulation containing *T. harzianum* T(MP)-7 and *B. amyloliquefaciens* B(TI)-3 as an effective biocontrol strategy against *P. cinnamomi* in avocado cultivation. The bioformulation supported high microbial populations and retained viability over time, while GC-MS analysis identified multiple antifungal metabolites contributing to pathogen suppression. Additionally, molecular analysis confirmed the presence of antifungal

CRedit authorship contribution statement

M. Ayyandurai: Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **G.S. Madhu:** Writing – review & editing, Writing

– original draft, Supervision. **K. Manonmani:** Supervision, Resources. **N. Rajinimala:** Visualization, Supervision, Software. **P. Mahalakshmi:** Supervision, Resources, Investigation, Data curation. **M. Theradimani:** Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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